BELLOMASSI

DIAGRAMS AND VISUAL DOCUMENTATION

Diagram Catalogue and Visual Documentation

Bellomassi Order-to-Delivery Management Platform

Document ID BM-DGM-001

Document control

Field	Value
Document title	Diagram Catalogue and Visual Documentation
Document ID	BM-DGM-001
Version	1.0
Status	Draft — for client review
Category	Diagrams and Visual Documentation
Related module(s)	All modules
Document owner	Solution Architect
Prepared by	Project documentation team (AI-assisted draft prepared from source-code analysis)
Reviewed by	*Pending — not yet reviewed*
Approval	*Not approved — awaiting client review (no approval has been given)*
Created date	2026-11-12
Last updated date	2026-11-12

> **Status labels used in this document:** VERIFIED (confirmed in the current source code or existing documents) · PROTOTYPE (works in the React/LocalStorage prototype with mock or simulated behaviour) · PROPOSED (recommended production design, not built) · PENDING CONFIRMATION (needs a client decision) · NOT IMPLEMENTED · NOT VERIFIED.

Purpose

Provide all required architecture, process, data and operations diagrams in one place with editable sources.

Scope

23 diagrams (system architecture, business process, workflows, lifecycle, roles, RBAC, ER, data flow, deployment, network, authentication, integration, CI/CD, backup and recovery).

Definitions

Other terms: see **BM-GLO-001** (Project Glossary).

References

- **BM-ARC-003** (Architecture Diagrams)
- **BM-ARC-001** (High-Level Design (HLD))
- **BM-WF-00** (Business Workflow Catalogue)

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1. Catalogue

Editable sources are Mermaid (`.mmd`) files; previews are exported as SVG and PNG in this folder. Diagrams never show a production component as existing: labels state VERIFIED (prototype) or PROPOSED (target).

ID	Diagram	Label	Files	Source
DGM-01	Overall System Architecture (prototype and target)	VERIFIED (prototype) / PROPOSED (target)	`mermaid/DGM-01_Overall_System_Architecture_prototype_and_target.mmd` · `svg/DGM-01_Overall_System_Architecture_prototype_and_target.svg` · `png/DGM-01_Overall_System_Architecture_prototype_and_target.png`	this catalogue
DGM-02	Business Process Architecture	VERIFIED	`mermaid/DGM-02_Business_Process_Architecture.mmd` · `svg/DGM-02_Business_Process_Architecture.svg` · `png/DGM-02_Business_Process_Architecture.png`	this catalogue

ID	Diagram	Label	Files	Source
DGM-03	Order-to-Delivery Flowchart	VERIFIED	`mermaid/DGM-03_Order_to_Delivery_Flowchart.mmd` · `svg/DGM-03_Order_to_Delivery_Flowchart.svg` · `png/DGM-03_Order_to_Delivery_Flowchart.png`	docs/workflow chapter 06
DGM-04	Order Status Lifecycle	VERIFIED	`mermaid/DGM-04_Order_Status_Lifecycle.mmd` · `svg/DGM-04_Order_Status_Lifecycle.svg` · `png/DGM-04_Order_Status_Lifecycle.png`	docs/workflow chapter 17
DGM-05	Approval Workflow	VERIFIED	`mermaid/DGM-05_Approval_Workflow.mmd` · `svg/DGM-05_Approval_Workflow.svg` · `png/DGM-05_Approval_Workflow.png`	docs/workflow chapter 07
DGM-06	Dispatch Workflow	VERIFIED	`mermaid/DGM-06_Dispatch_Workflow.mmd` · `svg/DGM-06_Dispatch_Workflow.svg` · `png/DGM-06_Dispatch_Workflow.png`	docs/workflow chapter 11
DGM-07	Driver Delivery Workflow	VERIFIED (GPS / OTP delivery simulated)	`mermaid/DGM-07_Driver_Delivery_Workflow.mmd` · `svg/DGM-07_Driver_Delivery_Workflow.svg` · `png/DGM-07_Driver_Delivery_Workflow.png`	docs/workflow chapter 08
DGM-08	Customer Order Workflow	VERIFIED (customer app scope PENDING CONFIRMATION)	`mermaid/DGM-08_Customer_Order_Workflow.mmd` · `svg/DGM-08_Customer_Order_Workflow.svg` · `png/DGM-08_Customer_Order_Workflow.png`	docs/workflow chapter 09
DGM-09	Role Interaction Diagram	VERIFIED	`mermaid/DGM-09_Role_Interaction_Diagram.mmd` · `svg/DGM-09_Role_Interaction_Diagram.svg` · `png/DGM-09_Role_Interaction_Diagram.png`	docs/workflow chapter 04
DGM-10	RBAC Diagram	VERIFIED (prototype) / PROPOSED (server enforcement)	`mermaid/DGM-10_RBAC_Diagram.mmd` · `svg/DGM-10_RBAC_Diagram.svg` · `png/DGM-10_RBAC_Diagram.png`	this catalogue
DGM-11	ER Diagram (overview)	PROPOSED	`mermaid/DGM-11_ER_Diagram_overview.mmd` · `svg/DGM-11_ER_Diagram_overview.svg` · `png/DGM-11_ER_Diagram_overview.png`	this catalogue
DGM-12	Data Flow Diagram	PROPOSED	`mermaid/DGM-12_Data_Flow_Diagram.mmd` · `svg/DGM-12_Data_Flow_Diagram.svg` · `png/DGM-12_Data_Flow_Diagram.png`	this catalogue
DGM-13	Deployment Architecture	PROPOSED	`mermaid/DGM-13_Deployment_Architecture.mmd` · `svg/DGM-13_Deployment_Architecture.svg` · `png/DGM-13_Deployment_Architecture.png`	this catalogue
DGM-14	Network Architecture	PROPOSED	`mermaid/DGM-14_Network_Architecture.mmd` · `svg/DGM-14_Network_Architecture.svg` · `png/DGM-14_Network_Architecture.png`	this catalogue
DGM-15	Authentication Flow (production)	PROPOSED	`mermaid/DGM-15_Authentication_Flow_production.mmd` · `svg/DGM-15_Authentication_Flow_production.svg` · `png/DGM-15_Authentication_Flow_production.png`	this catalogue
DGM-16	Integration Architecture	PROPOSED	`mermaid/DGM-16_Integration_Architecture.mmd` · `svg/DGM-16_Integration_Architecture.svg` · `png/DGM-16_Integration_Architecture.png`	this catalogue
DGM-17	CI/CD Pipeline	PROPOSED	`mermaid/DGM-17_CI_CD_Pipeline.mmd` · `svg/DGM-17_CI_CD_Pipeline.svg` · `png/DGM-17_CI_CD_Pipeline.png`	this catalogue
DGM-18	Backup and Recovery Architecture	PROPOSED	`mermaid/DGM-18_Backup_and_Recovery_Architecture.mmd` · `svg/DGM-18_Backup_and_Recovery_Architecture.svg` · `png/DGM-18_Backup_and_Recovery_Architecture.png`	this catalogue

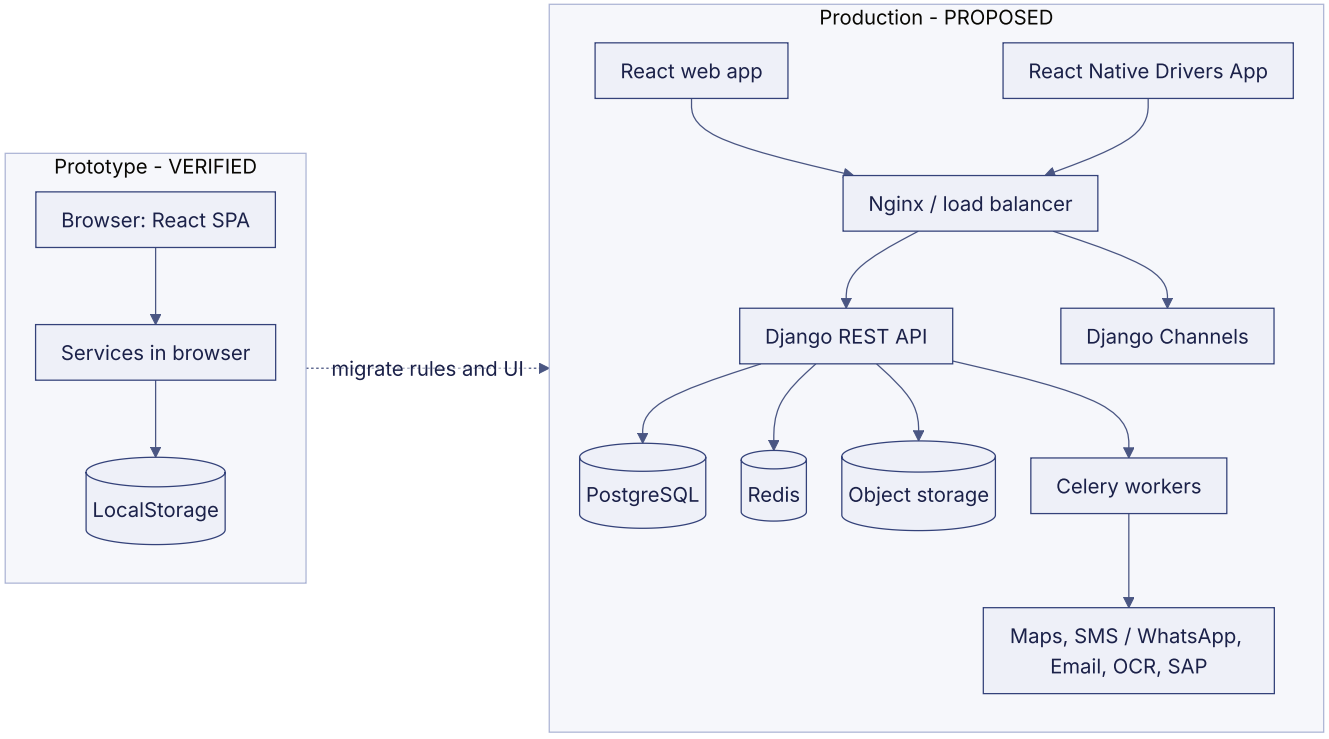
ID	Diagram	Label	Files	Source
DGM-19	System Context Diagram	VERIFIED (actors) / PROPOSED (external systems)	`mermaid/DGM-19_System_Context_Diagram.mmd` · `svg/DGM-19_System_Context_Diagram.svg` · `png/DGM-19_System_Context_Diagram.png`	this catalogue
DGM-20	Component Diagram (prototype)	VERIFIED	`mermaid/DGM-20_Component_Diagram_prototype.mmd` · `svg/DGM-20_Component_Diagram_prototype.svg` · `png/DGM-20_Component_Diagram_prototype.png`	this catalogue
DGM-21	Driver Delivery Architecture (production)	PROPOSED	`mermaid/DGM-21_Driver_Delivery_Architecture_production.mmd` · `svg/DGM-21_Driver_Delivery_Architecture_production.svg` · `png/DGM-21_Driver_Delivery_Architecture_production.png`	this catalogue
DGM-22	Customer Application Architecture	VERIFIED (prototype) / PROPOSED (production)	`mermaid/DGM-22_Customer_Application_Architecture.mmd` · `svg/DGM-22_Customer_Application_Architecture.svg` · `png/DGM-22_Customer_Application_Architecture.png`	this catalogue
DGM-23	Order Processing Architecture	VERIFIED (logic) / PROPOSED (server)	`mermaid/DGM-23_Order_Processing_Architecture.mmd` · `svg/DGM-23_Order_Processing_Architecture.svg` · `png/DGM-23_Order_Processing_Architecture.png`	this catalogue

2. Editing

Edit the `.mmd` file and re-render with Mermaid CLI: `mmdc -i mermaid/<file>.mmd -o svg/<file>.svg`. The workflow diagrams (DGM-03 … DGM-09) are maintained in `docs/workflow/` and copied here.

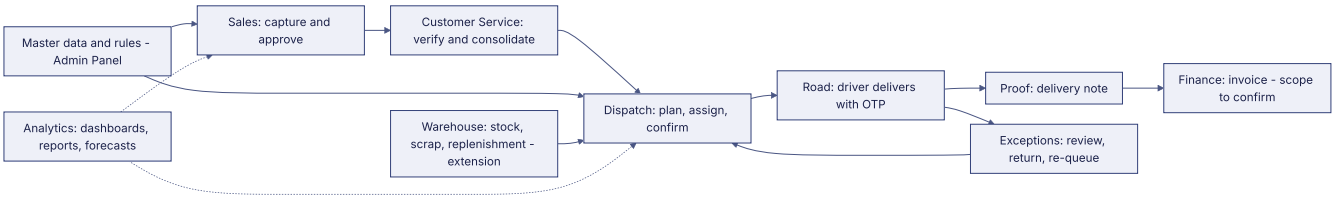
DGM-01 — Overall System Architecture (prototype and target)

Label: **VERIFIED (prototype) / PROPOSED (target)**



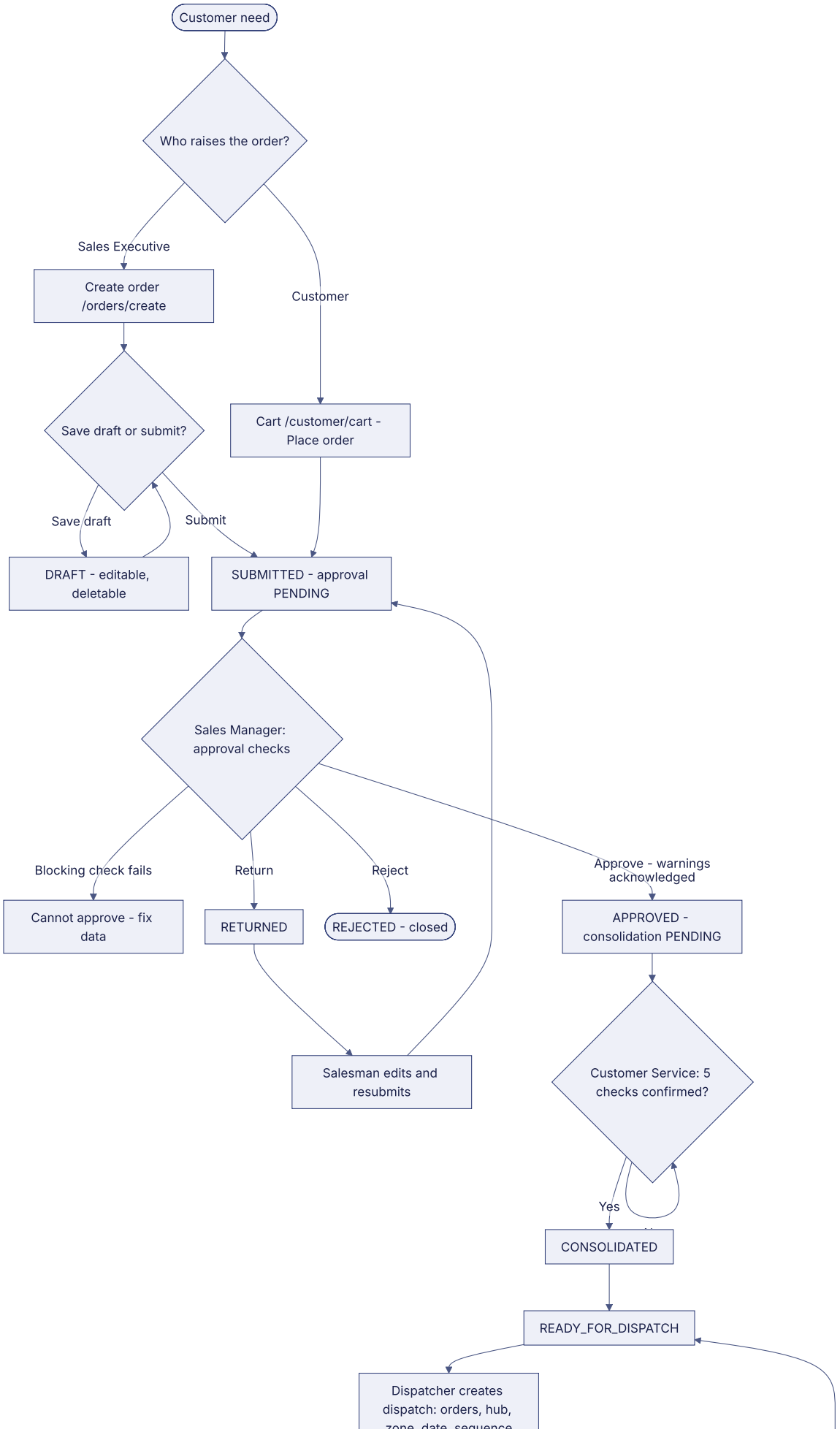
DGM-02 — Business Process Architecture

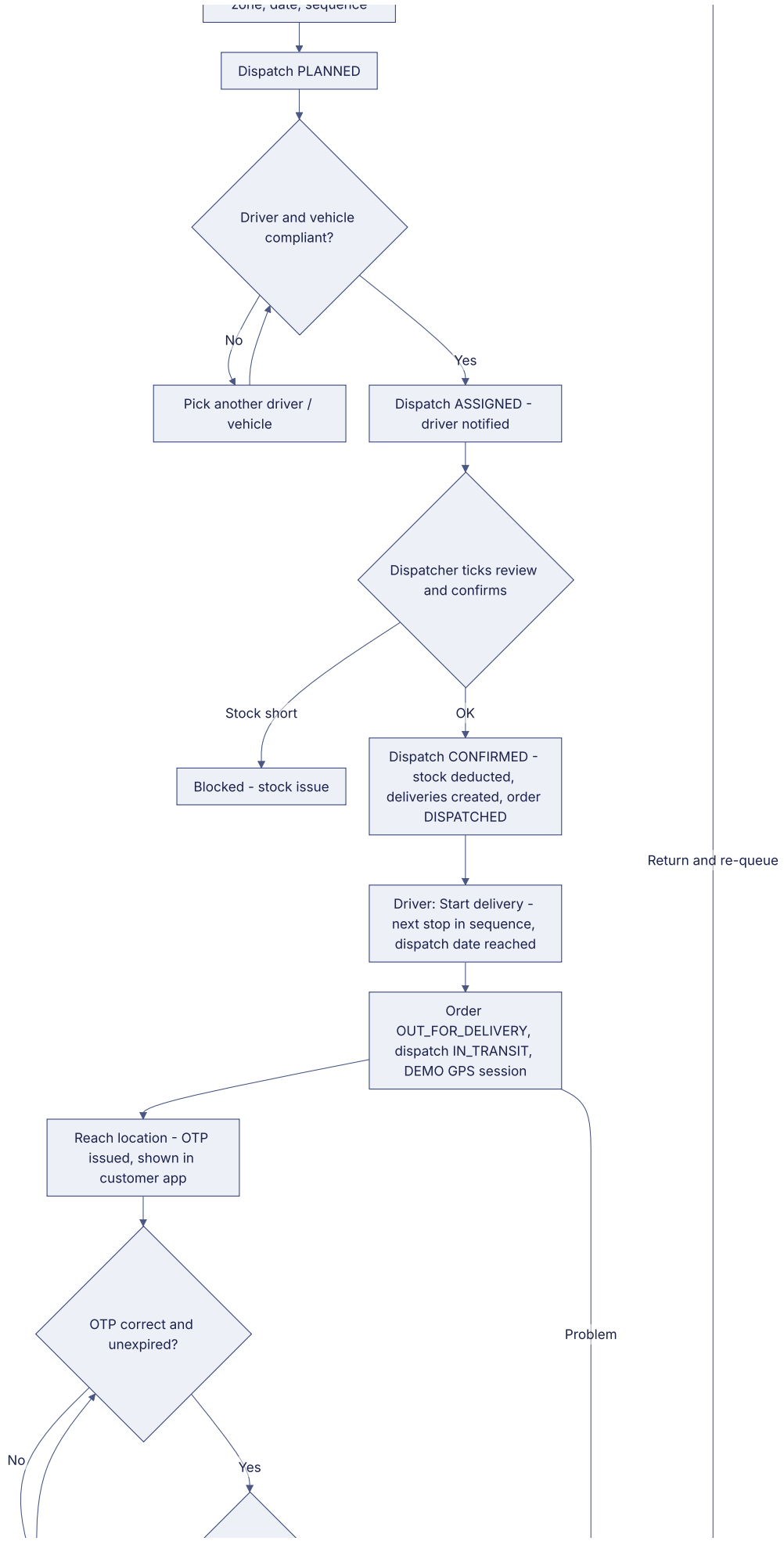
Label: **VERIFIED**

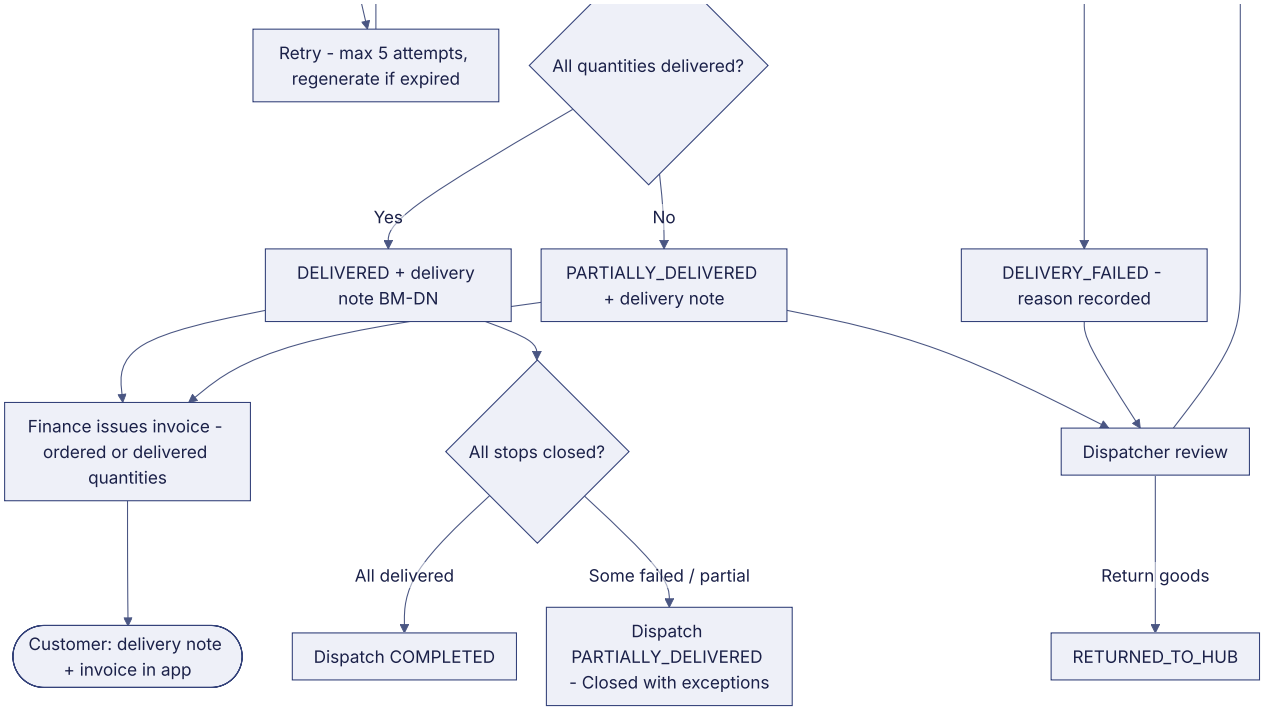


DGM-03 — Order-to-Delivery Flowchart

Label: **VERIFIED**

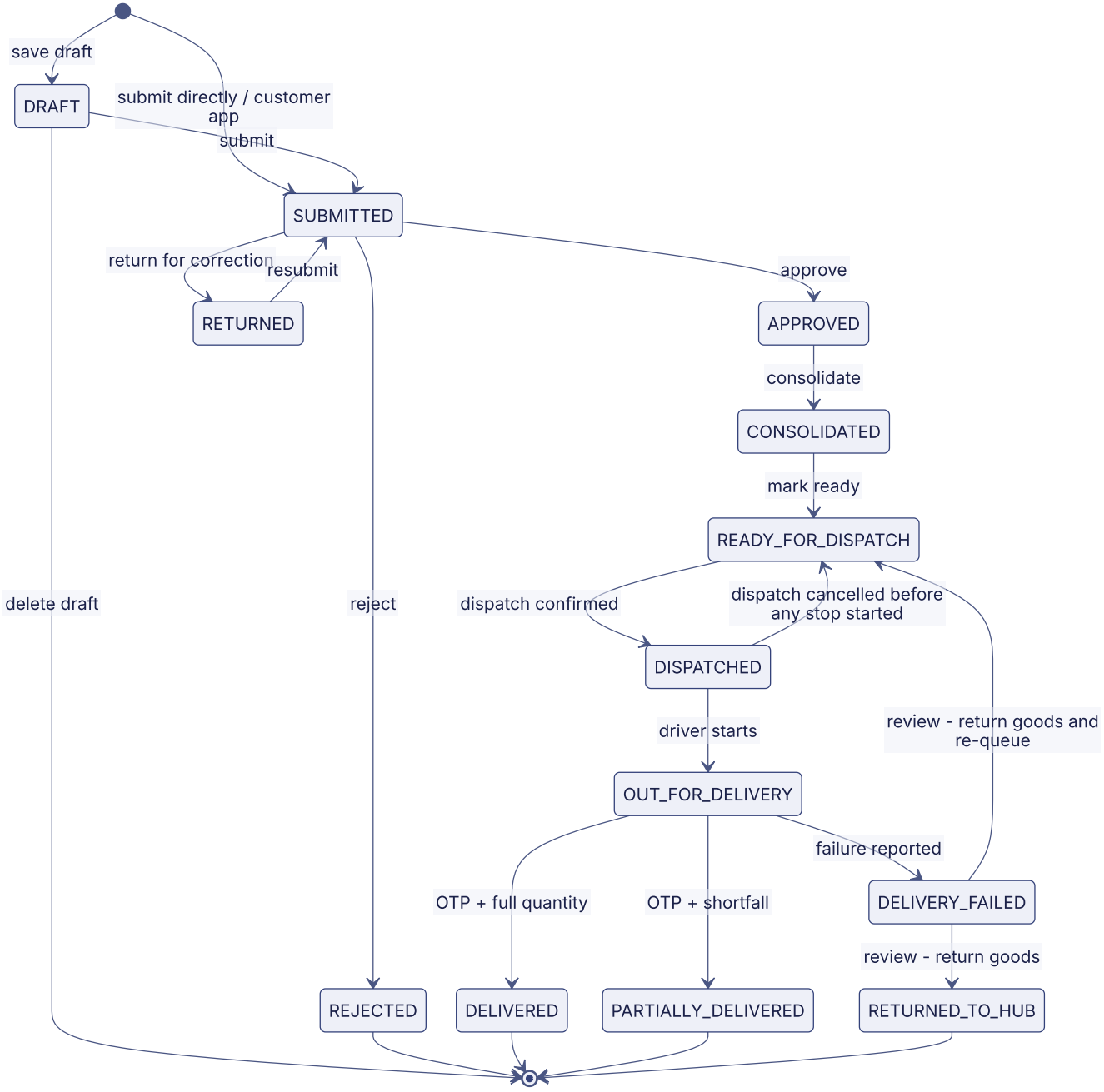

Customer need
Who raises the order?
Sales Executive
Customer
Create order /orders/create
Save draft or submit?
Cart /customer/cart - Place order
Save draft
Submit
DRAFT - editable, deletable
SUBMITTED - approval PENDING
Sales Manager: approval checks
Blocking check fails
Return
Reject
Approve - warnings acknowledged
Cannot approve - fix data
RETURNED
REJECTED - closed
APPROVED - consolidation PENDING
Salesman edits and resubmits
Customer Service: 5 checks confirmed?
Yes
CONSOLIDATED
READY_FOR_DISPATCH
Dispatcher creates dispatch: orders, hub,


Zone, date, sequence
Dispatch PLANNED
Driver and vehicle compliant?
No
Yes
Pick another driver / vehicle
Dispatch ASSIGNED - driver notified
Dispatcher ticks review and confirms
Stock short
OK
Blocked - stock issue
Dispatch CONFIRMED - stock deducted, deliveries created, order DISPATCHED
Return and re-queue
Driver: Start delivery - next stop in sequence, dispatch date reached
Order OUT_FOR_DELIVERY, dispatch IN_TRANSIT, DEMO GPS session
Reach location - OTP issued, shown in customer app
OTP correct and unexpired?
Problem
No
Yes



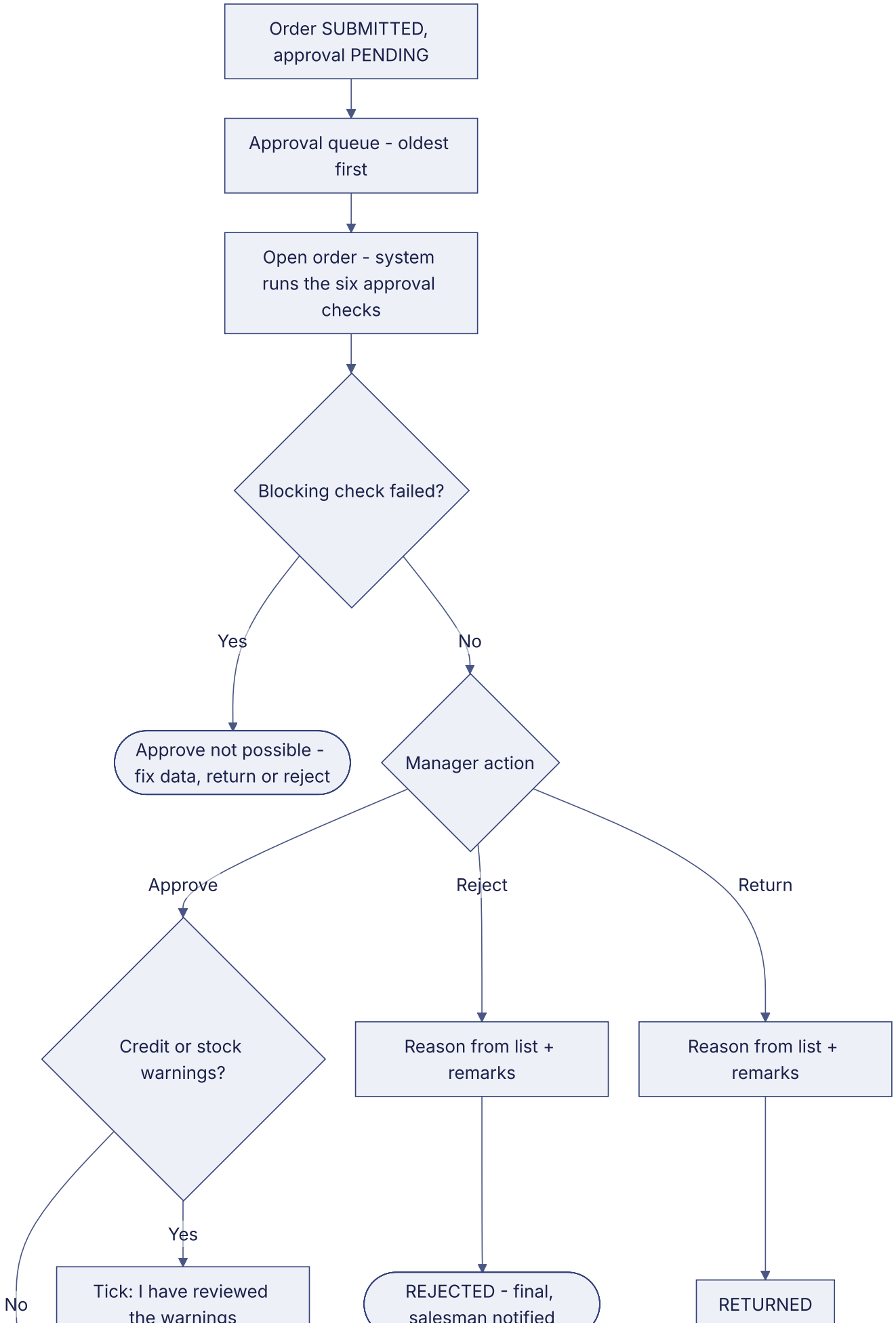
DGM-04 — Order Status Lifecycle

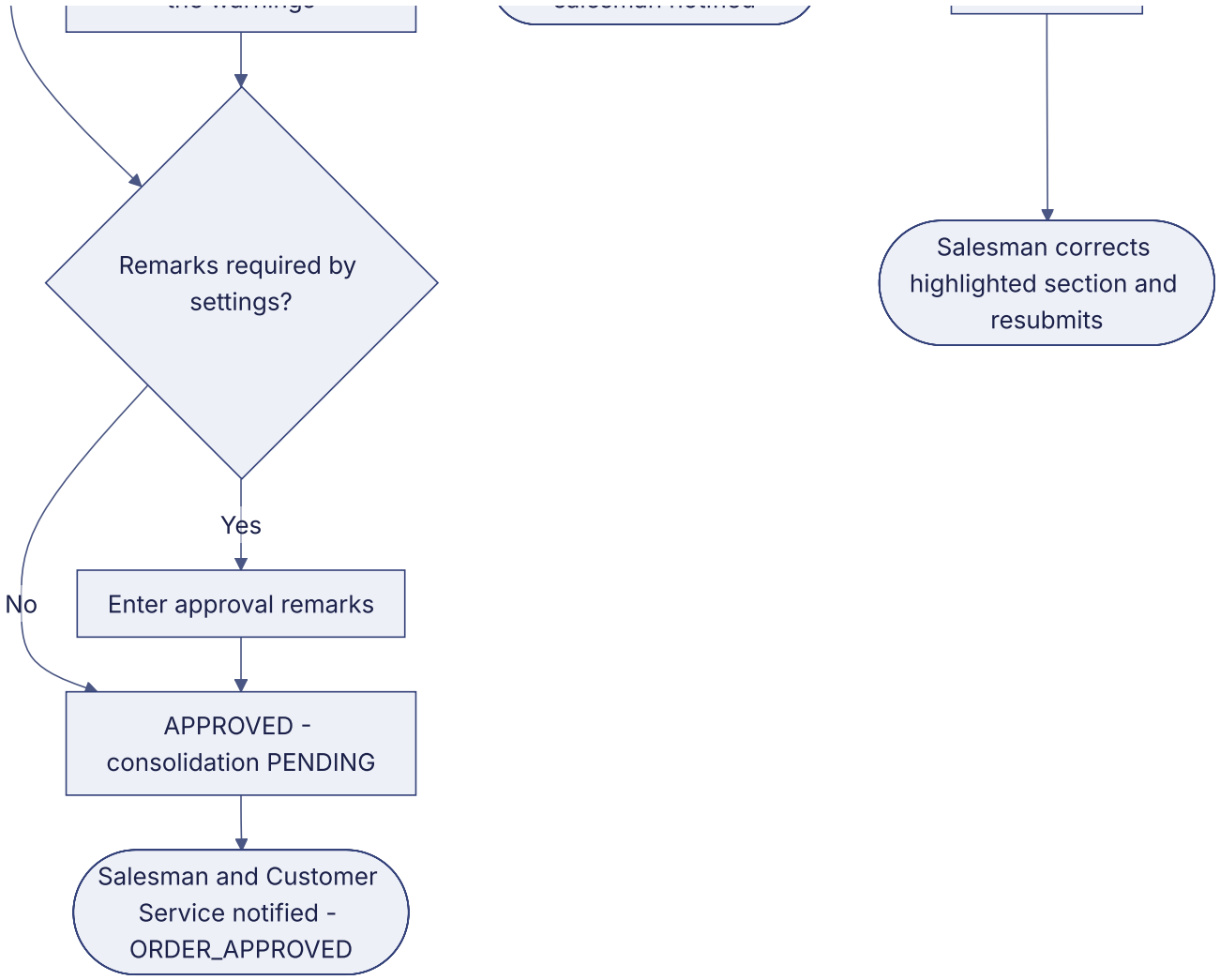
Label: **VERIFIED**



DGM-05 — Approval Workflow

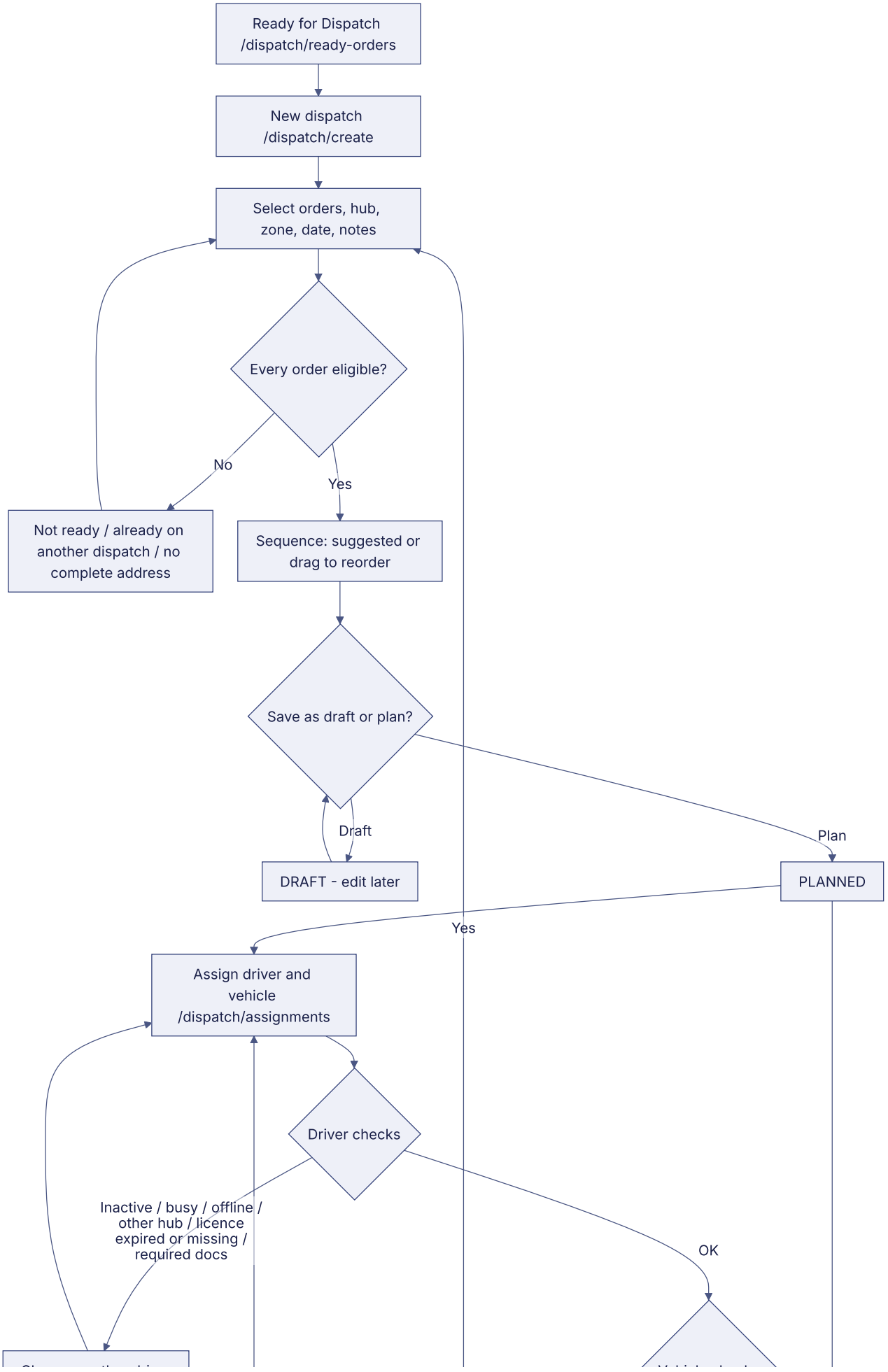
Label: **VERIFIED**

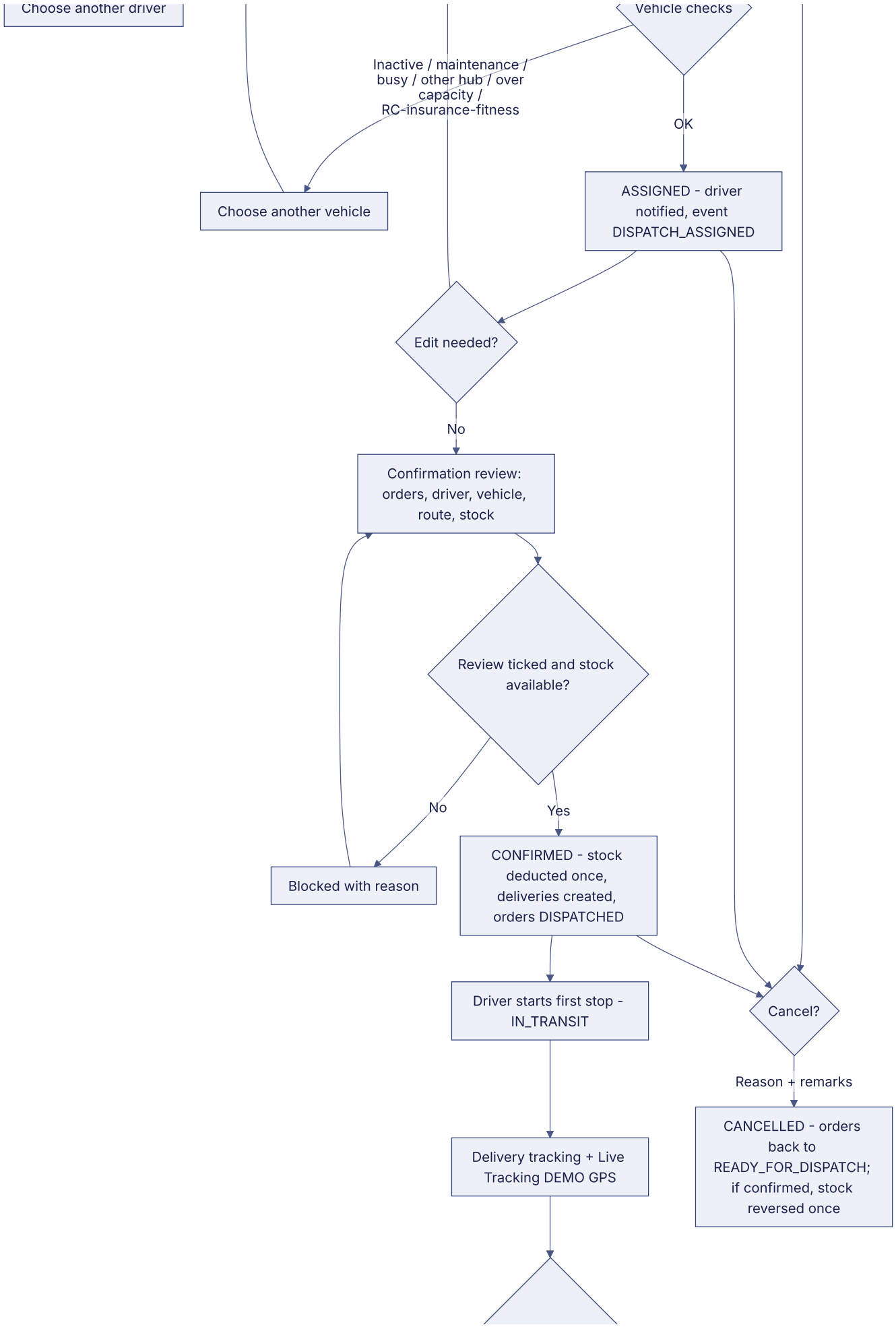

Order SUBMITTED, approval PENDING
Approval queue - oldest first
Open order - system runs the six approval checks
Blocking check failed?
Yes
No
Approve not possible - fix data, return or reject
Manager action
Approve
Reject
Return
Credit or stock warnings?
Reason from list + remarks
Reason from list + remarks
Yes
No
Tick: I have reviewed the warnings
REJECTED - final, salesman notified
RETURNED

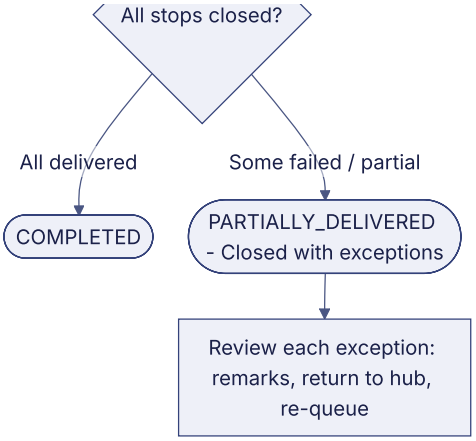


DGM-06 — Dispatch Workflow

Label: **VERIFIED**

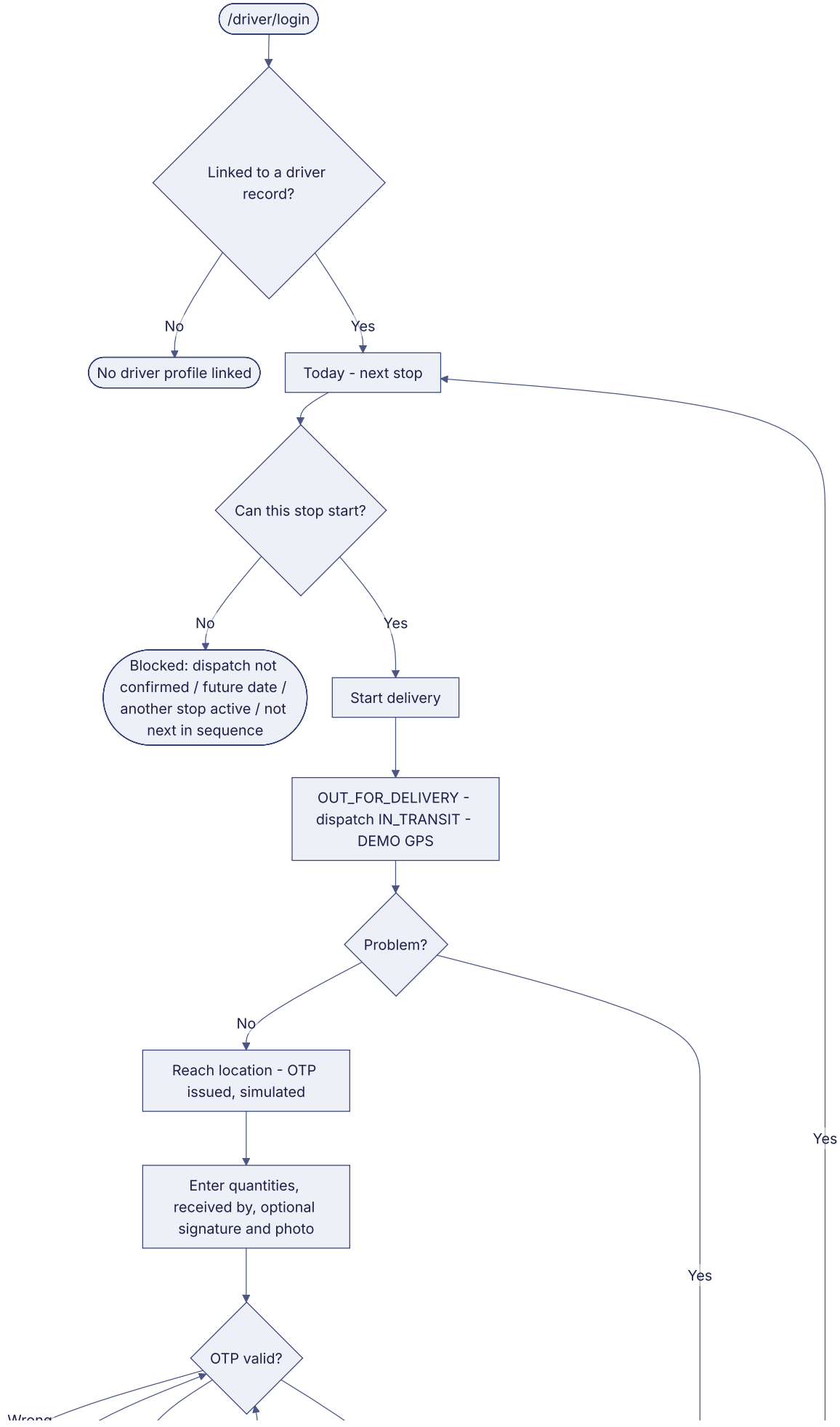

Ready for Dispatch
/dispatch/ready-orders
New dispatch
/dispatch/create
Select orders, hub,
zone, date, notes
Every order eligible?
No
Yes
Not ready / already on
another dispatch / no
complete address
Sequence: suggested or
drag to reorder
Save as draft or plan?
Draft
Plan
DRAFT - edit later
PLANNED
Yes
Assign driver and
vehicle
/dispatch/assignments
Driver checks
Inactive / busy / offline /
other hub / licence
expired or missing /
required docs
OK

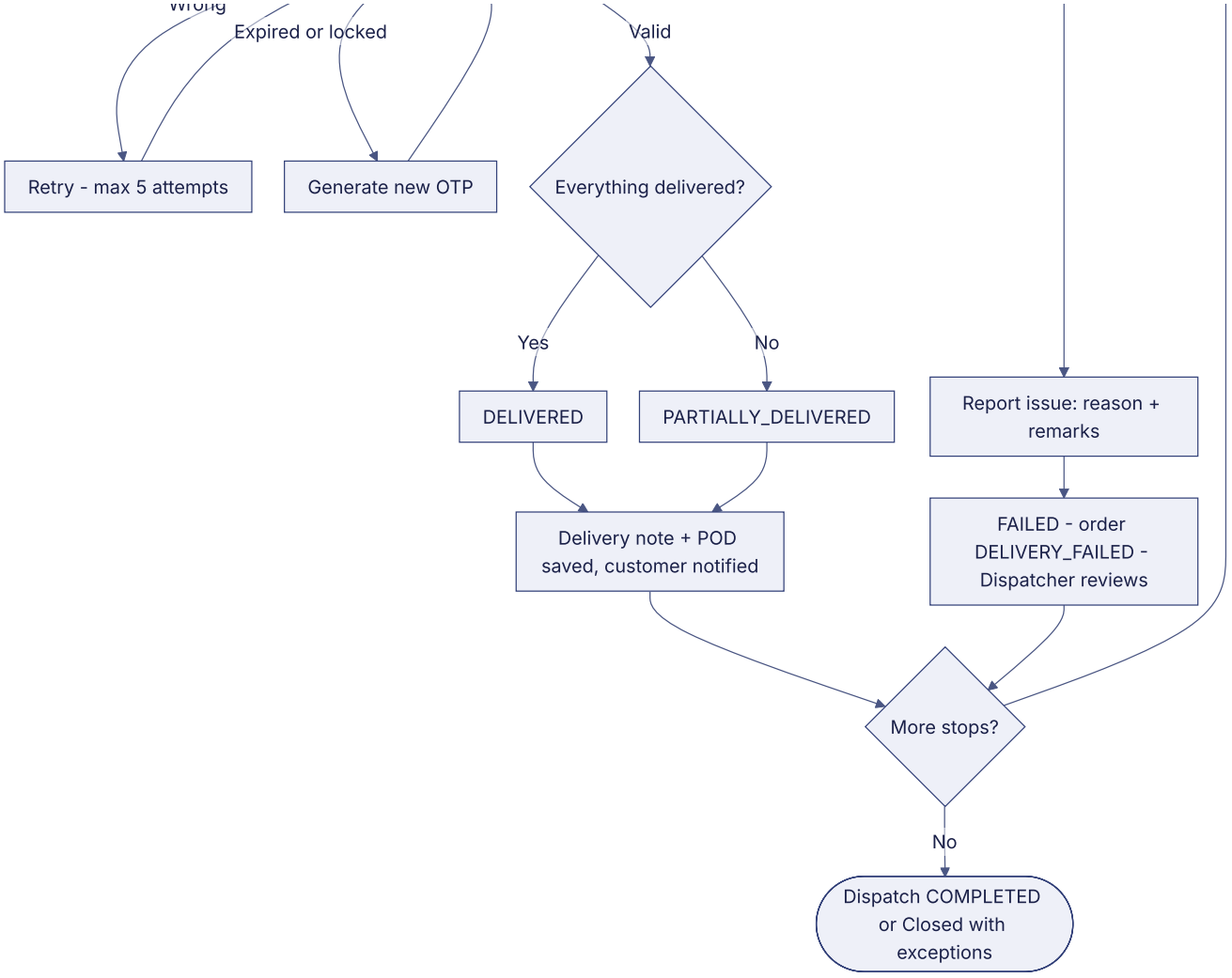

Choose another driver
Vehicle checks
Inactive / maintenance / busy / other hub / over capacity / RC-insurance-fitness
OK
Choose another vehicle
ASSIGNED - driver notified, event DISPATCH_ASSIGNED
Edit needed?
No
Confirmation review: orders, driver, vehicle, route, stock
Review ticked and stock available?
No
Yes
Blocked with reason
CONFIRMED - stock deducted once, deliveries created, orders DISPATCHED
Driver starts first stop - IN_TRANSIT
Cancel?
Reason + remarks
CANCELLED - orders back to READY_FOR_DISPATCH; if confirmed, stock reversed once
Delivery tracking + Live Tracking DEMO GPS



DGM-07 — Driver Delivery Workflow

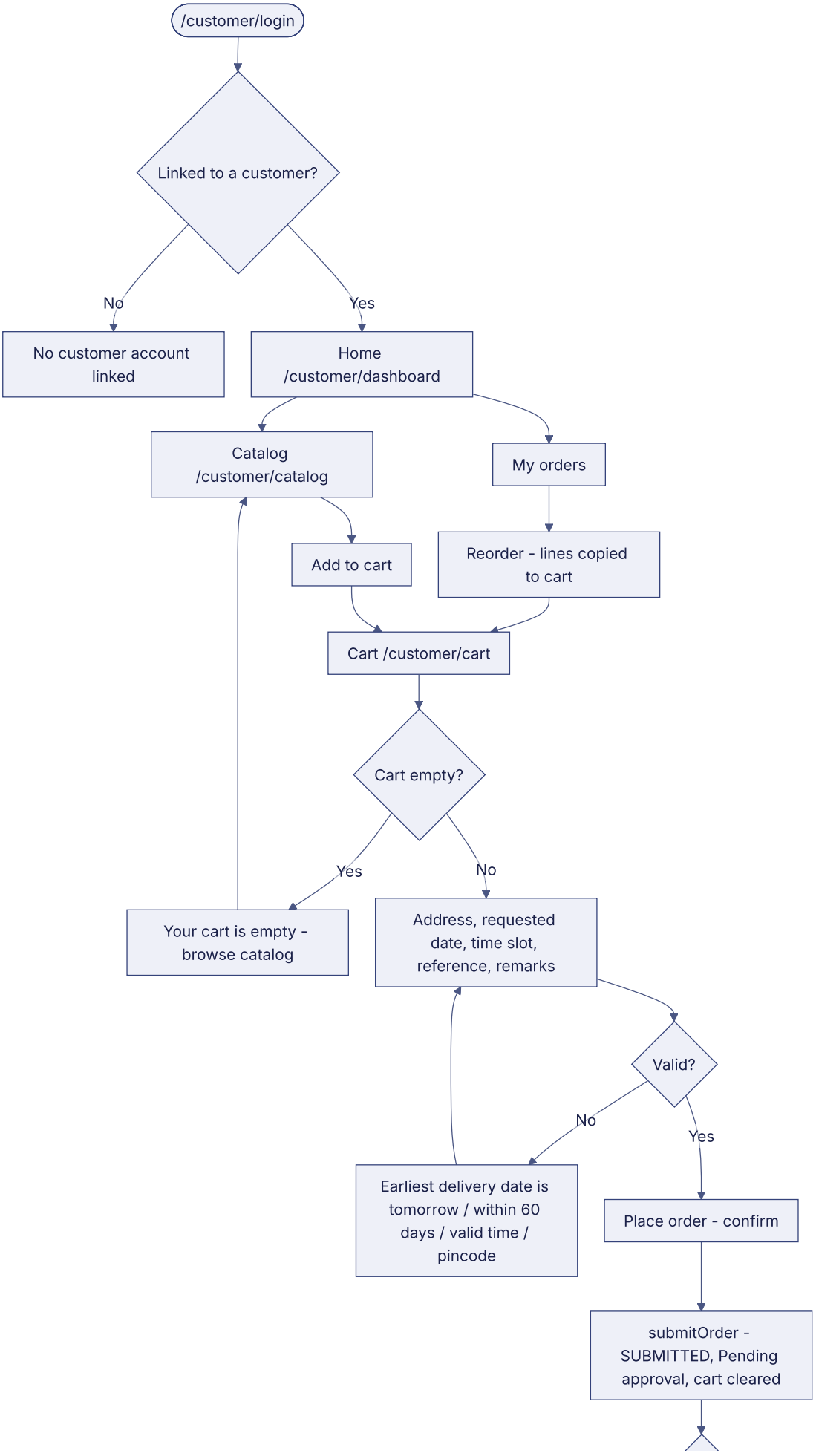
Label: **VERIFIED (GPS / OTP delivery simulated)**

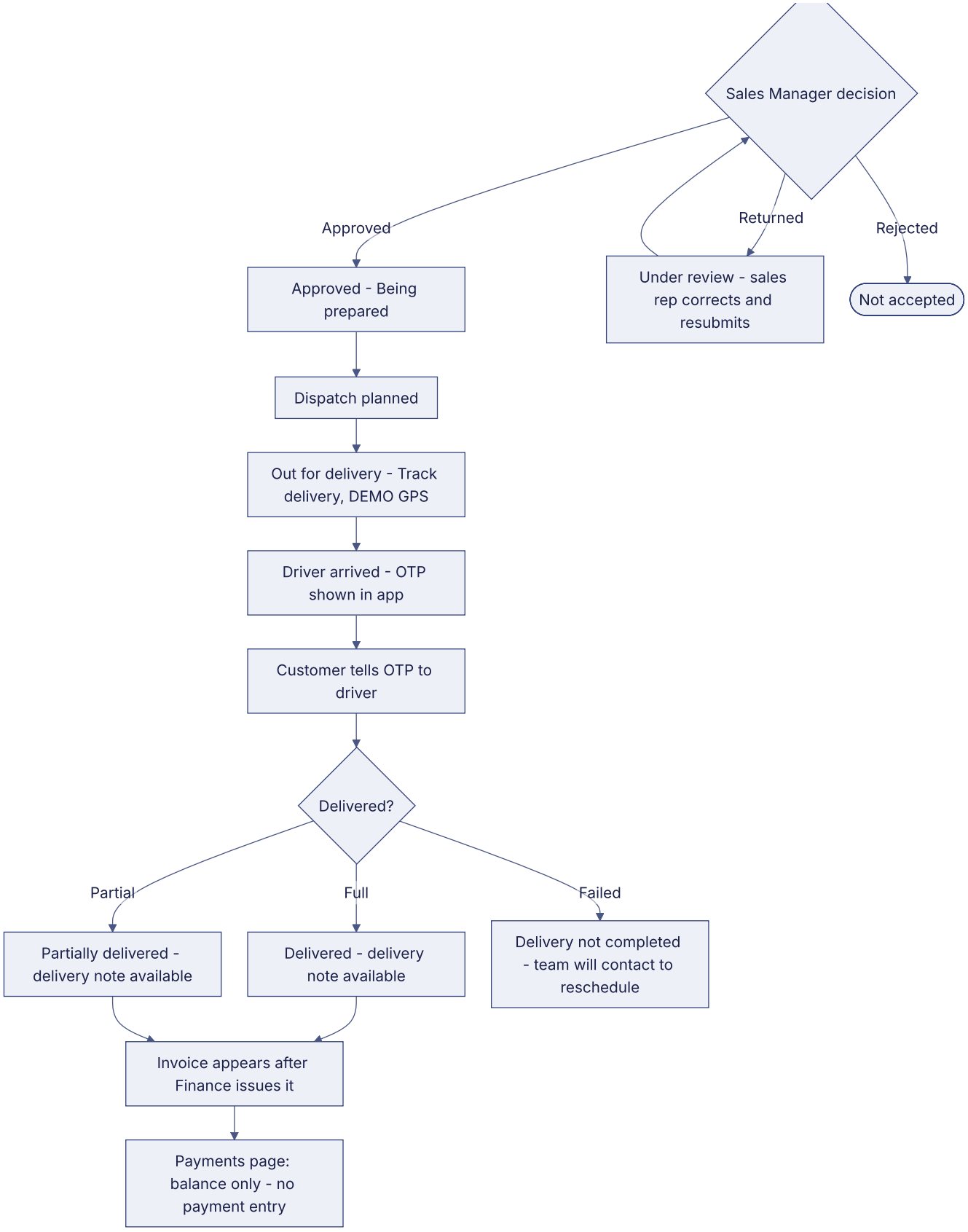

/driver/login
Linked to a driver record?
No
Yes
No driver profile linked
Today - next stop
Can this stop start?
No
Yes
Blocked: dispatch not confirmed / future date / another stop active / not next in sequence
Start delivery
OUT_FOR_DELIVERY - dispatch IN_TRANSIT - DEMO GPS
Problem?
No
Reach location - OTP issued, simulated
Enter quantities, received by, optional signature and photo
OTP valid?
Yes
Yes



DGM-08 — Customer Order Workflow

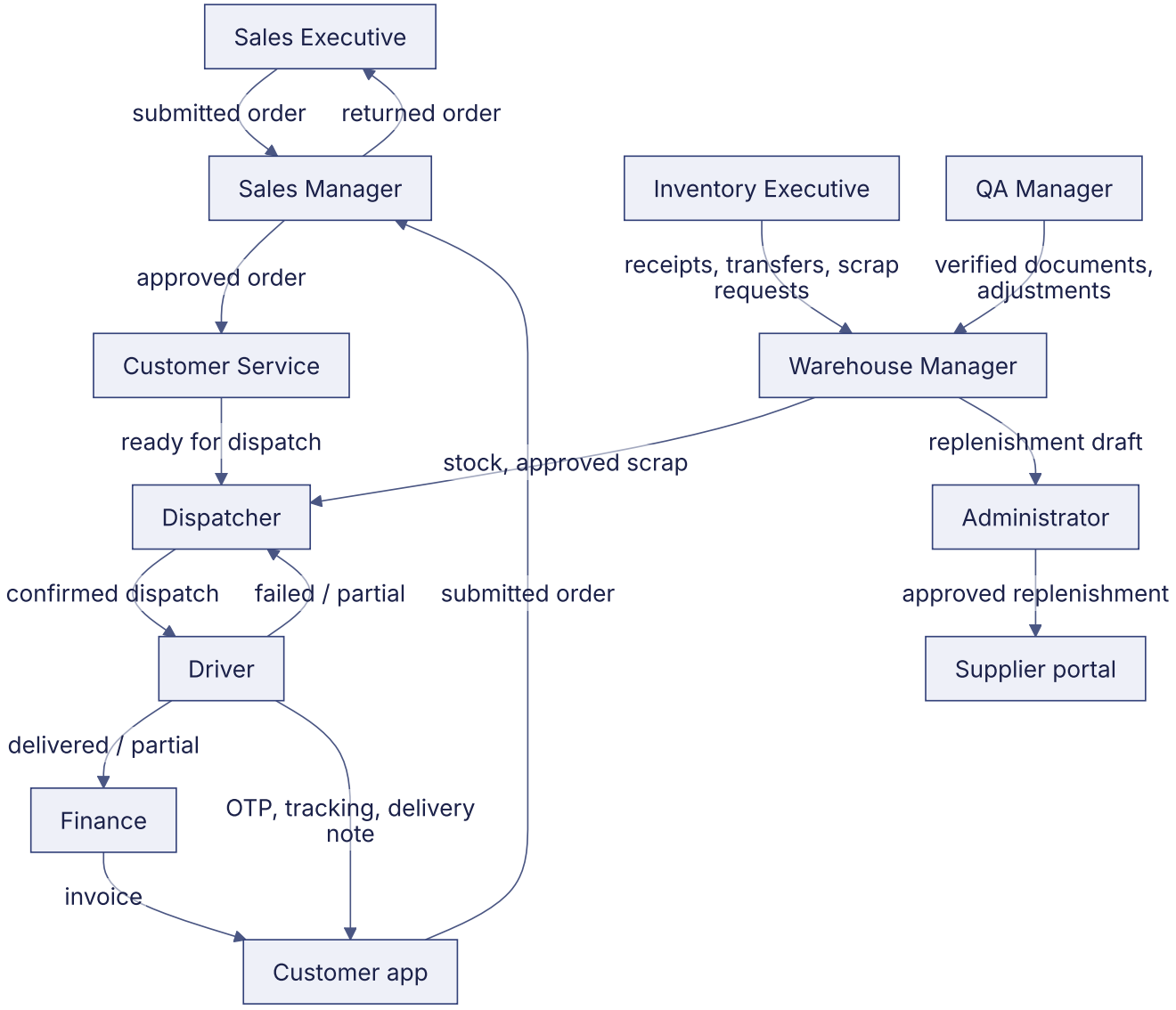
Label: **VERIFIED (customer app scope PENDING CONFIRMATION)**


/customer/login
Linked to a customer?
No
Yes
No customer account linked
Home /customer/dashboard
Catalog /customer/catalog
My orders
Add to cart
Reorder - lines copied to cart
Cart /customer/cart
Cart empty?
Yes
No
Your cart is empty - browse catalog
Address, requested date, time slot, reference, remarks
Valid?
No
Yes
Earliest delivery date is tomorrow / within 60 days / valid time / pincode
Place order - confirm
submitOrder - SUBMITTED, Pending approval, cart cleared



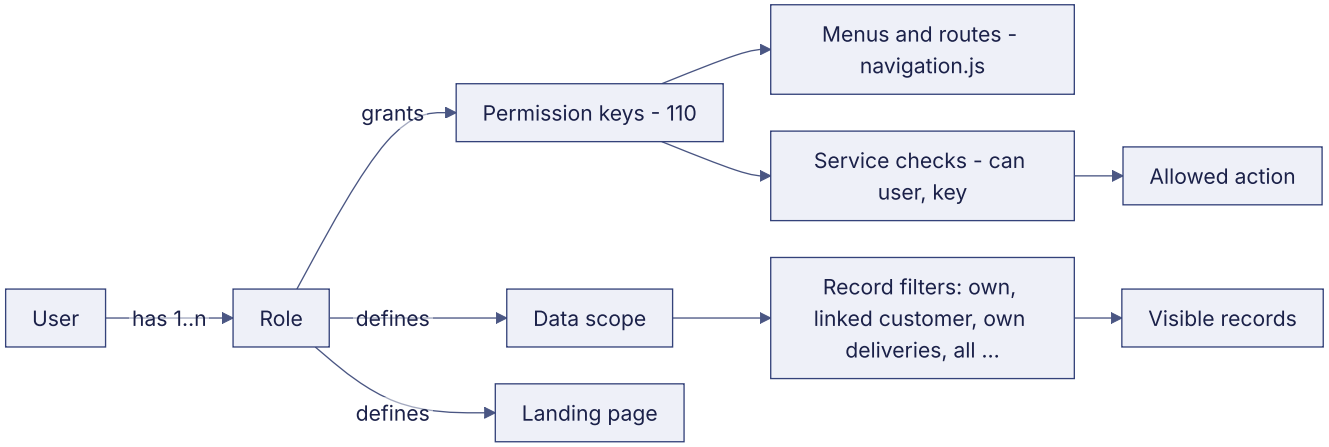
DGM-09 — Role Interaction Diagram

Label: **VERIFIED**



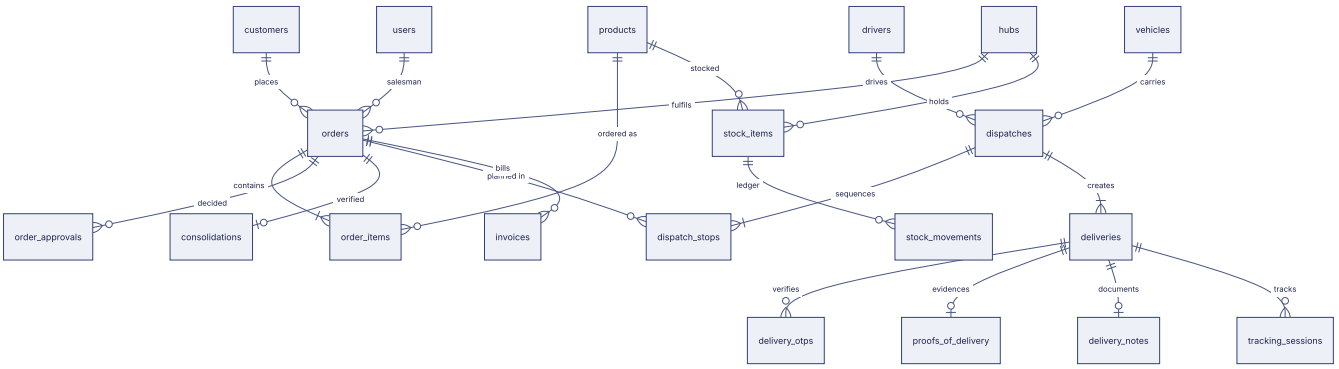
DGM-10 — RBAC Diagram

Label: **VERIFIED (prototype) / PROPOSED (server enforcement)**



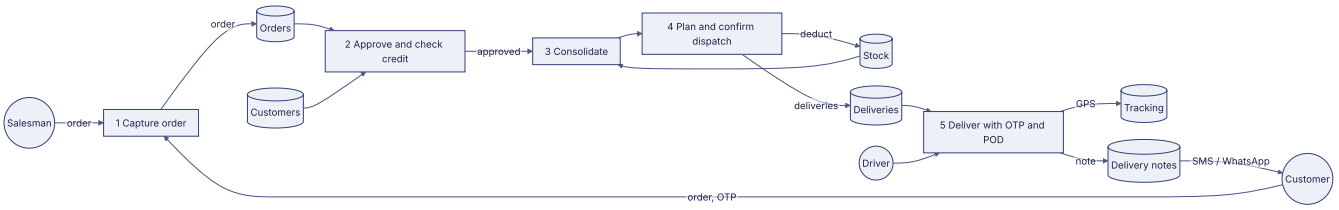
DGM-11 — ER Diagram (overview)

Label: **PROPOSED**



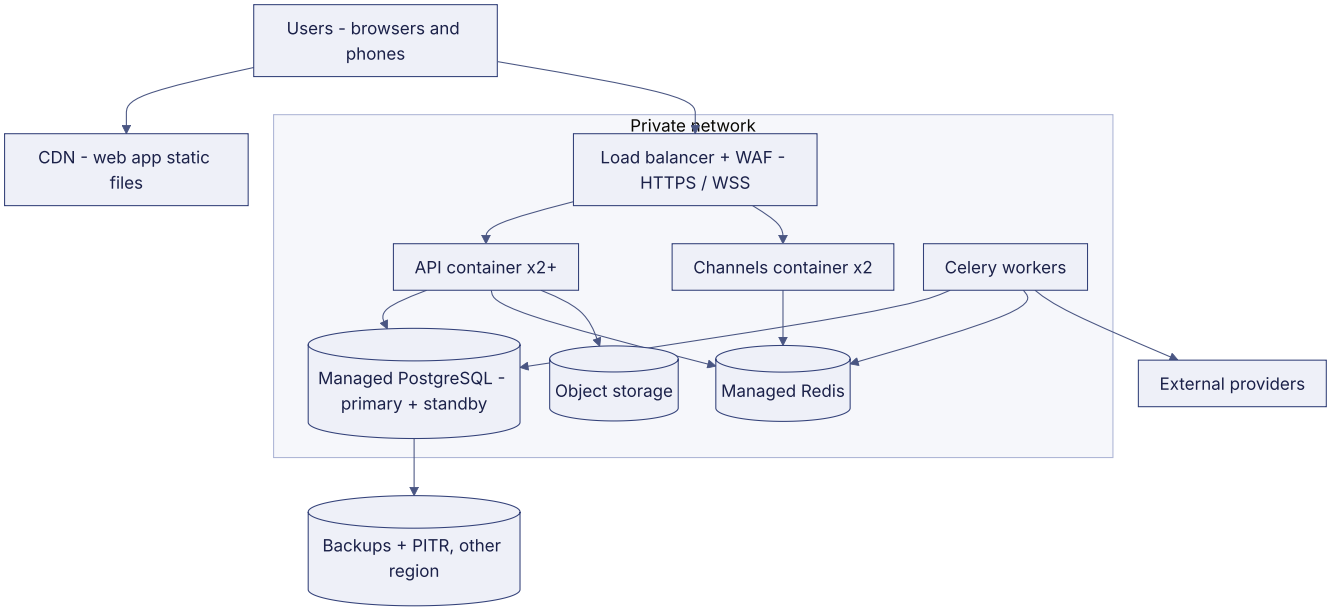
DGM-12 — Data Flow Diagram

Label: **PROPOSED**



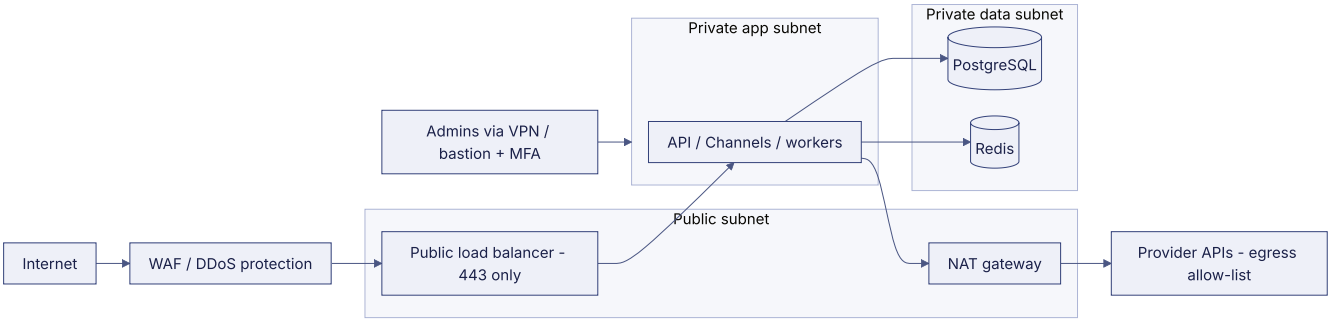
DGM-13 — Deployment Architecture

Label: **PROPOSED**



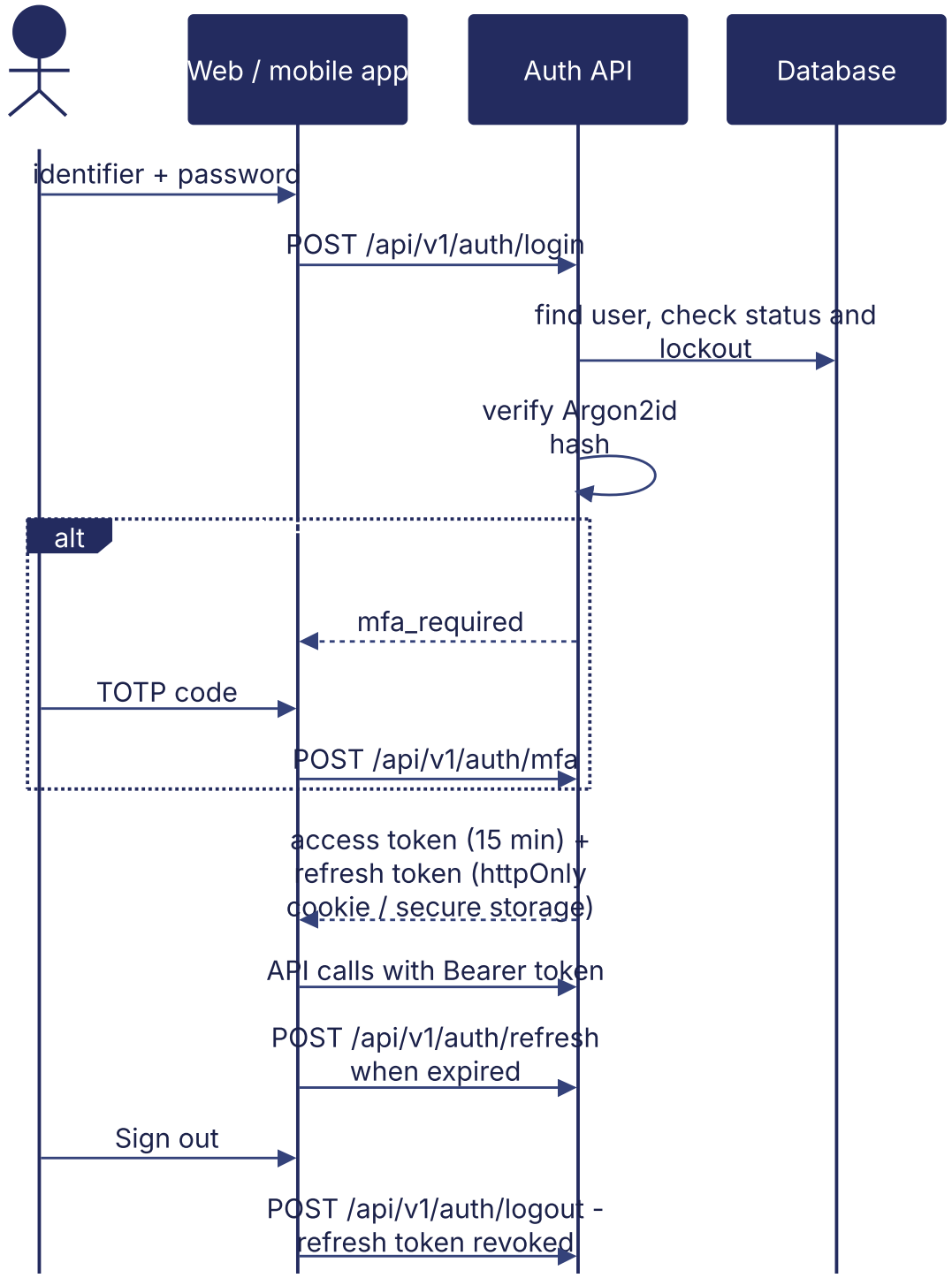
DGM-14 — Network Architecture

Label: **PROPOSED**



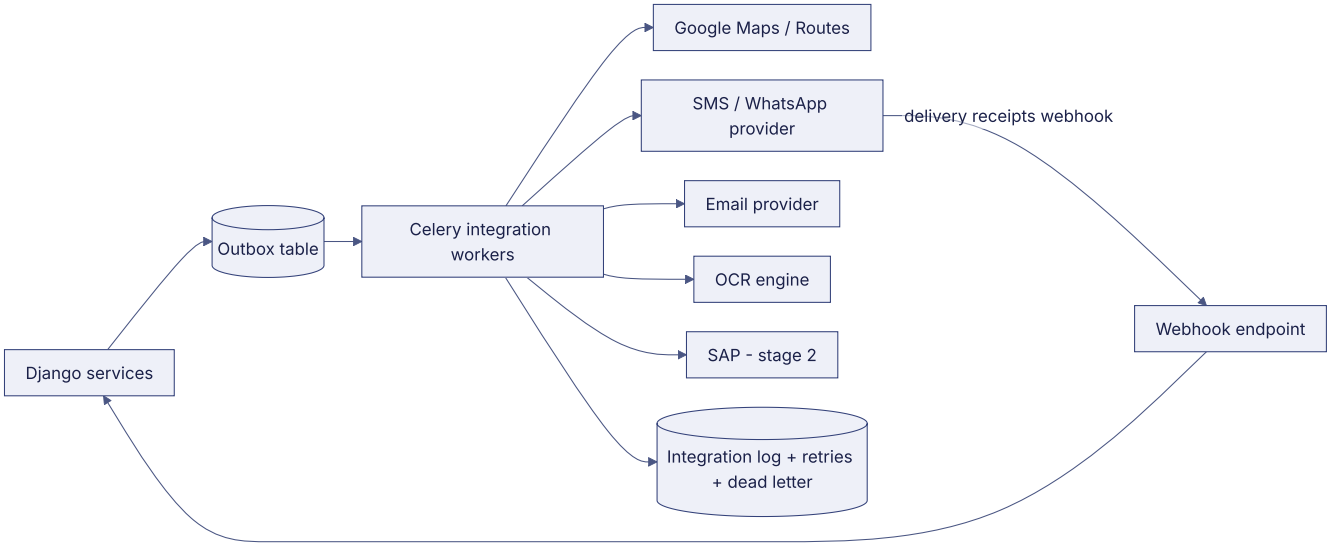
DGM-15 — Authentication Flow (production)

Label: **PROPOSED**



DGM-16 — Integration Architecture

Label: **PROPOSED**



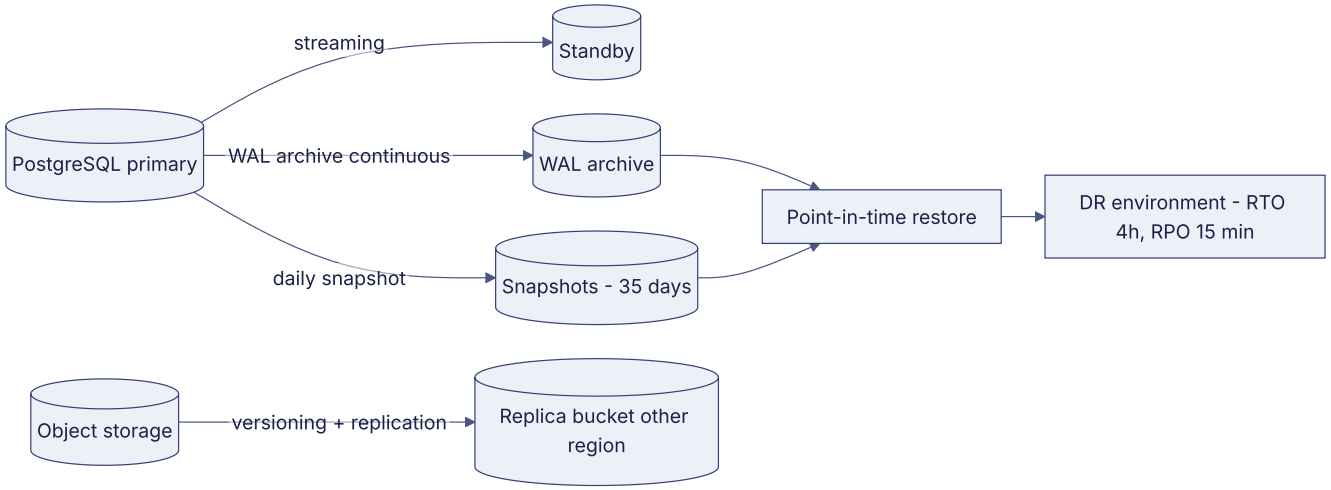
DGM-17 — CI/CD Pipeline

Label: **PROPOSED**



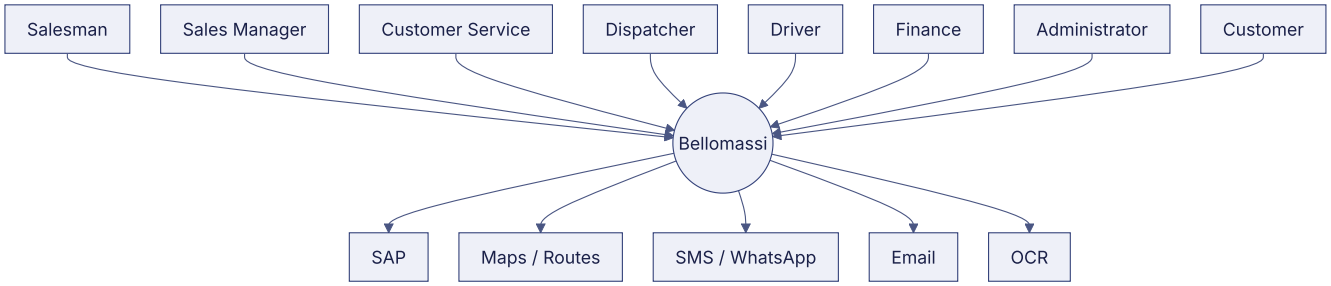
DGM-18 — Backup and Recovery Architecture

Label: **PROPOSED**



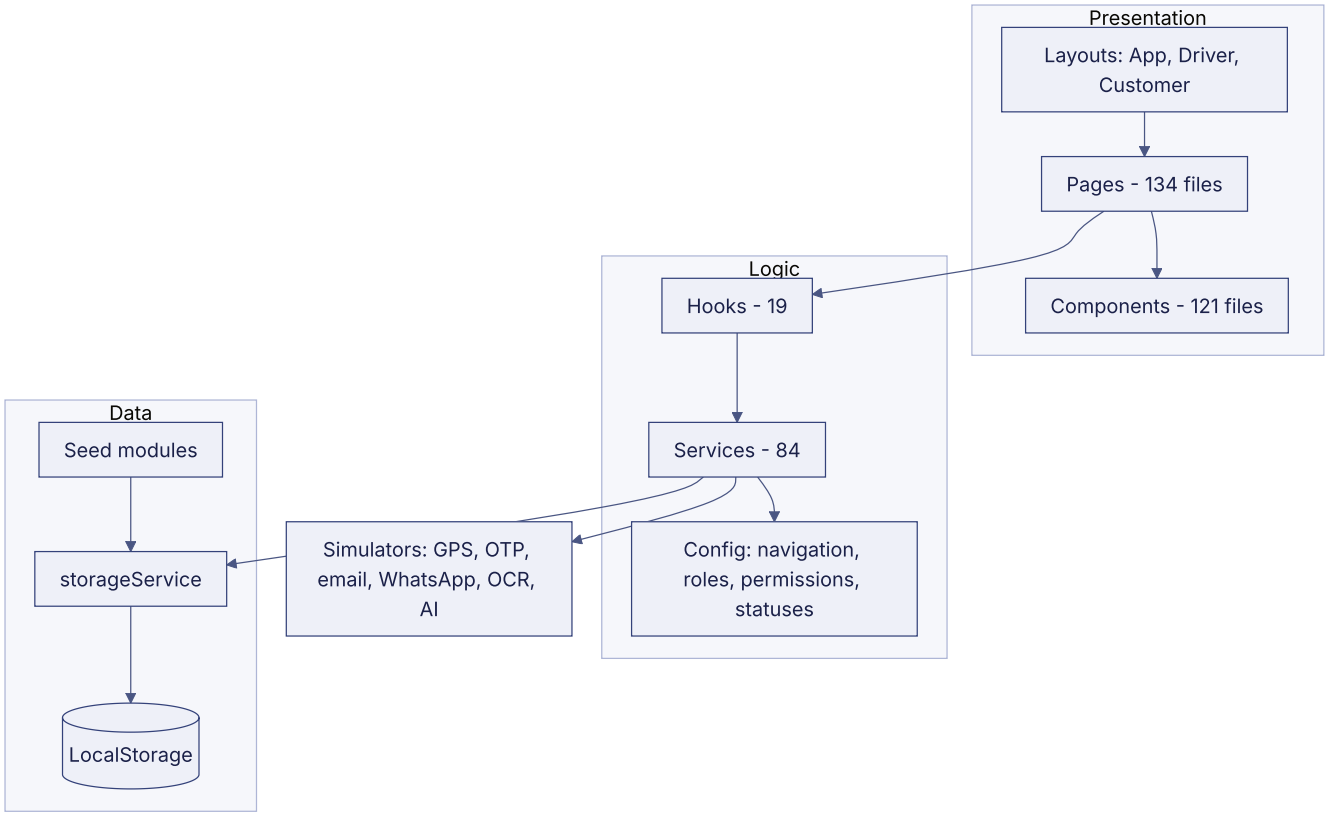
DGM-19 — System Context Diagram

Label: **VERIFIED (actors) / PROPOSED (external systems)**



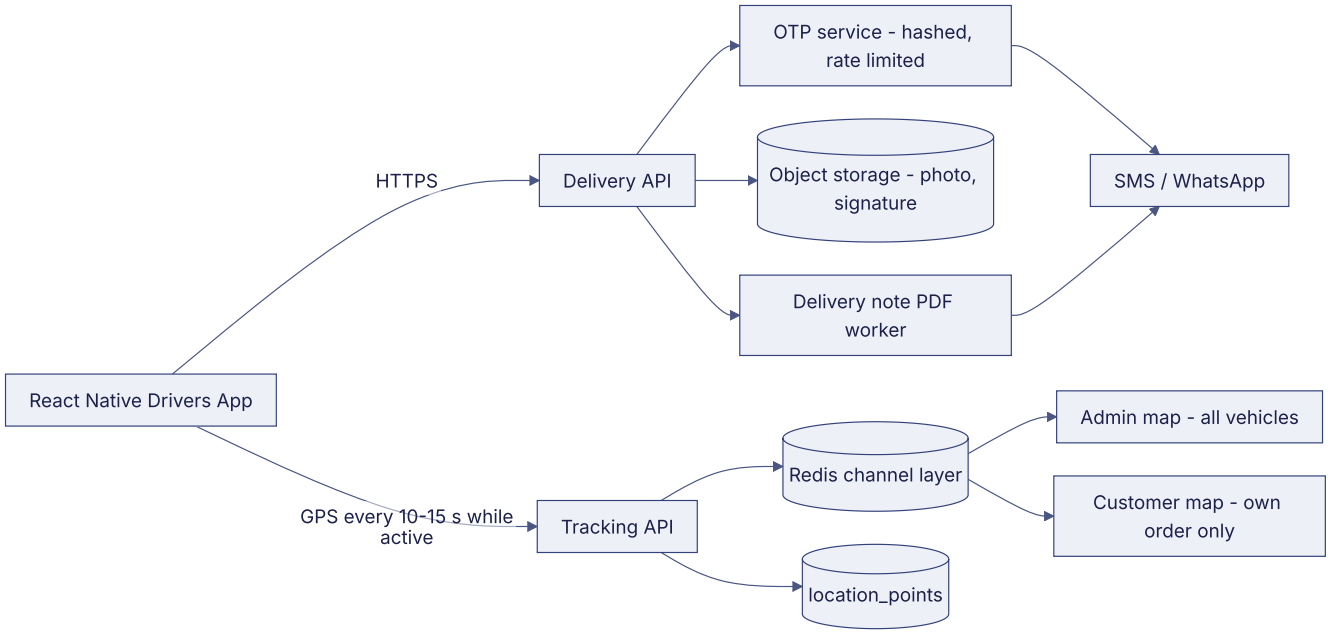
DGM-20 — Component Diagram (prototype)

Label: **VERIFIED**



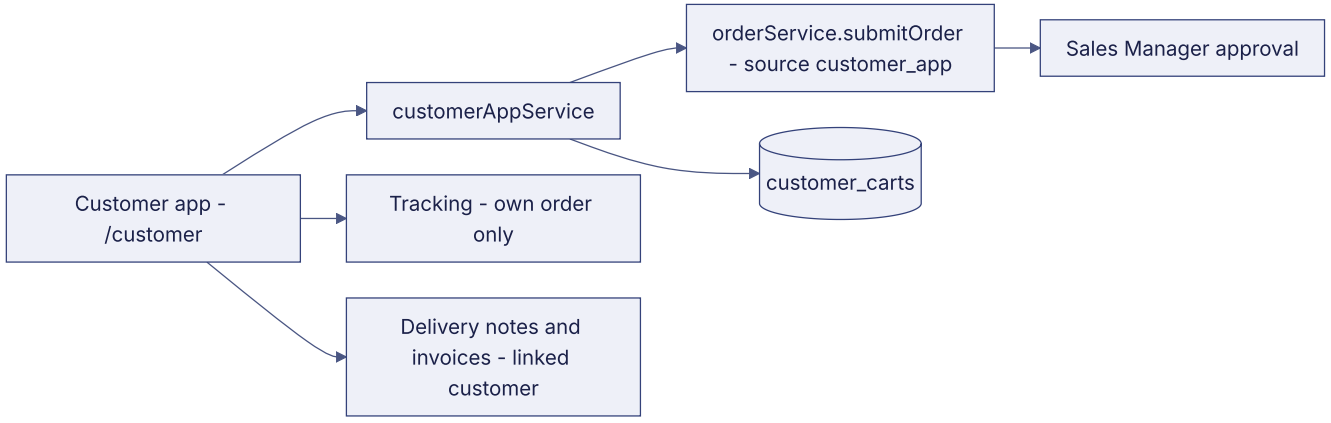
DGM-21 — Driver Delivery Architecture (production)

Label: **PROPOSED**



DGM-22 — Customer Application Architecture

Label: **VERIFIED (prototype) / PROPOSED (production)**



DGM-23 — Order Processing Architecture

Label: **VERIFIED (logic) / PROPOSED (server)**



Revision history

Version	Date	Author	Description
1.0	2026-11-12	Documentation team	Initial draft prepared from analysis of the bellomassi-premium-ui source code and existing project documents.